

What is <ctype.h>

The <ctype.h> header file in the C Standard Library provides a set of functions for character classification and transformation. These functions are particularly useful for testing whether a character belongs to a specific category (such as alphabetic, numeric, or whitespace) and for converting characters between uppercase and lowercase.

Key Features of <ctype.h>

- **Character Classification Functions:** These functions return non-zero values (true) if the input character satisfies the specified condition and zero (false) otherwise. Examples include:
 - `isalnum()`: Checks if the character is alphanumeric (a letter or digit) ^[1] ^[2].
 - `isalpha()`: Checks if the character is an alphabetic letter ^[1] ^[3].
 - `isdigit()`: Checks if the character is a decimal digit ^[1] ^[2].
 - `isspace()`: Checks if the character is a whitespace character like space, tab, or newline ^[3] ^[2].
 - `ispunct()`: Checks if the character is a punctuation mark ^[3] ^[2].
- **Character Transformation Functions:**
 - `toupper()`: Converts a lowercase letter to uppercase ^[2] ^[4].
 - `tolower()`: Converts an uppercase letter to lowercase ^[2] ^[4].

Syntax

To use <ctype.h>, include it in your program:

```
#include <ctype.h>
```

Example Usage

Here's an example demonstrating the use of `isalnum()`:

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char ch;
    printf("Enter a character: ");
    scanf("%c", &ch);
```

```

    if (isalnum(ch)) {
        printf("%c is an alphanumeric character.\n", ch);
    } else {
        printf("%c is not an alphanumeric character.\n", ch);
    }

    return 0;
}

```

Common Functions in <ctype.h>

Function	Description
isalnum()	Checks if the character is alphanumeric
isalpha()	Checks if the character is alphabetic
isdigit()	Checks if the character is a decimal digit
isspace()	Checks if the character is whitespace
ispunct()	Checks if the character is punctuation
toupper()	Converts a lowercase letter to uppercase
tolower()	Converts an uppercase letter to lowercase

These functions simplify tasks involving text processing by allowing developers to classify and manipulate characters efficiently.



1. <https://www.includehelp.com/c-library-functions/ctype-example-programs.aspx>
2. https://www.tutorialspoint.com/c_standard_library/ctype_h.htm
3. https://www.w3schools.com/c/c_ref_ctype.php
4. <https://www.w3resource.com/c-programming/ctype/index.php>